Supplementary file 3
Results from pairwise network meta-analysis and network network meta-analysis

Data are the OR or MD (95% CrI) in the column-defining treatment compared with the row-defining treatment. With treatment as the boundary, the lower left part of the table is the result of network network meta-analysis, and the upper right part of the table is the result of pairwise network meta-analysis. For network network meta-analysis, OR lower than 1 or MD lower than 0 favour the column-defining treatment. For pairwise network meta-analysis, OR higher than 1 or MD higher than 0 favour the row-defining treatment. To obtain OR or MD for comparisons in the opposite direction, reciprocals should be taken. Significant results are in bold and underscored. OR:odds ratio. MD:mean difference. CrI:credible interval. EPN: enteral nutrition and/or adjuvant peripheral parenteral nutrition. TPN: total parenteral nutrition.

Table S 3.1 Pairwise and network estimated odds ratio (with 95% CrI) of interventions on nosocomial infection

Synbiotics	--	6.60 (0.73, 60.49)	**5.20 (2.10, 14.1)**	--
0.24 (0.08, 0.83)	**Probiotics**	--	1.20 (0.66, 2.9)	--
0.16 (0.05, 0.52)	0.65 (0.23, 1.90)	**Prebiotics**	0.80 (0.38, 2.03)	--
0.19 (0.08, 0.47)	0.82 (0.34, 1.50)	1.24 (0.49, 2.56)	**EPN**	1.50 (0.81, 3.01)
0.13 (0.04, 0.38)	0.55 (0.17, 1.26)	0.85 (0.25, 2.10)	0.69 (0.33, 1.23)	**TPN**

Table S 3.2 Pairwise and network estimated odds ratio (with 95% CrI) of interventions on bloodstream infection

Synbiotics	--	1.8e+04(1.9e-05, 2.7e+13)	2.10 (0.40, 12.39)	-
0.50 (0.11, 3.47)	**Probiotics**	--	1.20 (0.41, 10.52)	--
0.27 (0.04, 1.38)	0.53 (0.05, 2.50)	**Prebiotics**	0.68 (0.11, 3.90)	--
0.43 (0.11, 1.39)	0.87 (0.19, 2.08)	1.60 (0.43, 7.03)	**EPN**	1.50 (0.30, 10.54)
0.28 (0.04, 1.47)	0.57 (0.06, 2.36)	1.03 (0.16, 7.22)	0.66 (0.16, 2.21)	**TPN**

Table S 3.3 Pairwise and network estimated odds ratio (with 95% CrI) of interventions on urinary tract infections

Synbiotics	--	1.8e-06(1.0e-38, 1.8e+25)	3.20 (0.22, 100.05)	--
0.42 (0.04, 5.24)	**Probiotics**	--	1.40 (0.21, 26.77)	--
0.41 (0.04, 11.9)	0.95 (0.11, 22.81)	**Prebiotics**	1.20 (0.14, 420.12)	--
0.31 (0.04, 1.81)	0.75 (0.12, 3.00)	0.76 (0.04, 3.82)	**EPN**	1.70 (0.23, 83.27)
0.20 (0.01, 1.80)	0.47 (0.02, 3.34)	0.47 (0.01, 4.06)	0.63 (0.07, 2.83)	**TPN**

Table S 3.4 Pairwise and network estimated odds ratio (with 95% CrI) of interventions on diarrhea

Synbiotics	--	--	2.80 (0.80, 11.27)	--
0.60 (0.15, 3.13)	**Probiotics**	--	1.70 (0.89, 4.50)	--
7.87 (0.37, 370.18)	12.53 (0.66, 490.91)	**Prebiotics**	**20.79 (1.40, 980.73)**	--
0.36 (0.09, 1.25)	0.60 (0.22, 1.13)	**0.05 (0, 0.71)**	**EPN**	0.75 (0.14, 3.10)
0.50 (0.07, 3.77)	0.84 (0.13, 4.09)	0.06 (0, 1.51)	1.35 (0.32, 6.89)	**TPN**

Table S 3.5 Pairwise and network estimated odds ratio (with 95% CrI) of interventions on hospital mortality

Synbiotics	--	--	1.60 (0.96, 2.90)	--
0.81 (0.53, 1.23)	**Probiotics**	--	1.10 (0.83, 1.60)	--
1.22 (0.66, 2.30)	1.50 (0.89, 2.61)	**Prebiotics**	2.20 (1.10, 4.80)	--
0.76 (0.52, 1.09)	0.94 (0.76, 1.15)	0.63 (0.37, 1.01)	**EPN**	1.00 (0.75, 1.70)
0.76 (0.46, 1.15)	0.95 (0.64, 1.23)	0.62 (0.34, 1.04)	1.00 (0.74, 1.22)	**TPN**

Table S 3.6 Pairwise and network estimated odds ratio (with 95% CrI) of interventions on ICU mortality

Synbiotics	--	0.92 (0.04, 21.77)	1.50 (0.79, 2.80)	-
0.69 (0.34, 1.35)	**Probiotics**	-	1.00 (0.71, 1.30)	-
0.96 (0.36, 2.59)	1.41 (0.60, 3.33)	**Prebiotics**	0.80 (0.07, 9.02)	--
0.69 (0.38, 1.26)	1.00 (0.74, 1.41)	0.71 (0.32, 1.59)	**EPN**	0.98 (0.70, 1.70)
0.69 (0.30, 1.37)	1.01 (0.54, 1.63)	0.72 (0.27, 1.66)	1.02 (0.58, 1.44)	**TPN**

Table S 3.7 Pairwise and network estimated mean difference (with 95% CrI) of interventions on hospital lengtn of stay

Synbiotics	--	--	-2.10 (-10.29, 6.20)	--
3.90 (-4.73, 12.61)	**Probiotics**	--	1.90 (-0.95, 4.70)	--
2.05 (-8.41, 11.79)	-1.87 (-8.82, 4.50)	**Prebiotics**	-0.01 (-6.30, 5.70)	--
2.05 (-6.14, 10.37)	-1.86 (-4.69, 0.96)	0.03 (-5.67, 6.31)	**EPN**	0.95 (-2.40, 4.30)
1.12 (-7.83, 10.01)	-2.81 (-7.18, 1.63)	-0.89 (-7.59, 6.21)	-0.93 (-4.27, 2.46)	**TPN**

Table S 3.8 Pairwise and network estimated mean difference (with 95% CrI) of interventions on ICU length of stay

Synbiotics	--	1.60 (-9.10, 12.74)	3.80 (-1.40, 9.30)	--
-3.43 (-9.20, 2.10)	**Probiotics**	--	0.58 (-1.90, 3.10)	--
-1.86 (-8.92, 4.83)	1.54 (-4.37, 7.67)	**Prebiotics**	1.70 (-4.00, 7.50)	--
-4.00 (-9.35, 0.90)	-0.59 (-3.07, 1.92)	-2.14 (-7.70, 3.32)	**EPN**	1.70 (-2.04, 5.40)
-5.72 (-12.22, 0.30)	-2.29 (-6.72, 2.09)	-3.86 (-10.5, 2.72)	-1.7 (-5.49, 1.91)	**TPN**

Table S 3.9 Pairwise and network estimated mean difference (with 95% CrI) of interventions on the duration of mechanical ventilation

Synbiotics	--	-0.51 (-9.20, 8.10)	2.80 (-1.80, 7.90)	--
-1.27 (-6.76, 3.61)	**Probiotics**	--	1.50 (-0.93, 4.00)	--
-1.17 (-6.69, 4.37)	0.07 (-4.36, 5.03)	**Prebiotics**	1.10 (-2.80, 5.40)	--
-2.77 (-7.59, 1.48)	-1.48 (-3.97, 0.93)	-1.59 (-5.87, 2.14)	**EPN**	1.10 (-3.30, 5.40)
-3.78 (-10.26, 2.16)	-2.55 (-7.35, 2.44)	-2.63 (-8.65, 3.03)	-1.05 (-5.23, 3.32)	**TPN**